Modernizing .NET to Serverless

Native AOT and SnapStart, Live

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1097ms

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p50 cold start. ASP.NET Core API on AWS Lambda.

Today

320ms

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320ms

p50 cold start. ASP.NET Core API on AWS Lambda.

Two paths. Neither is flip-a-switch.

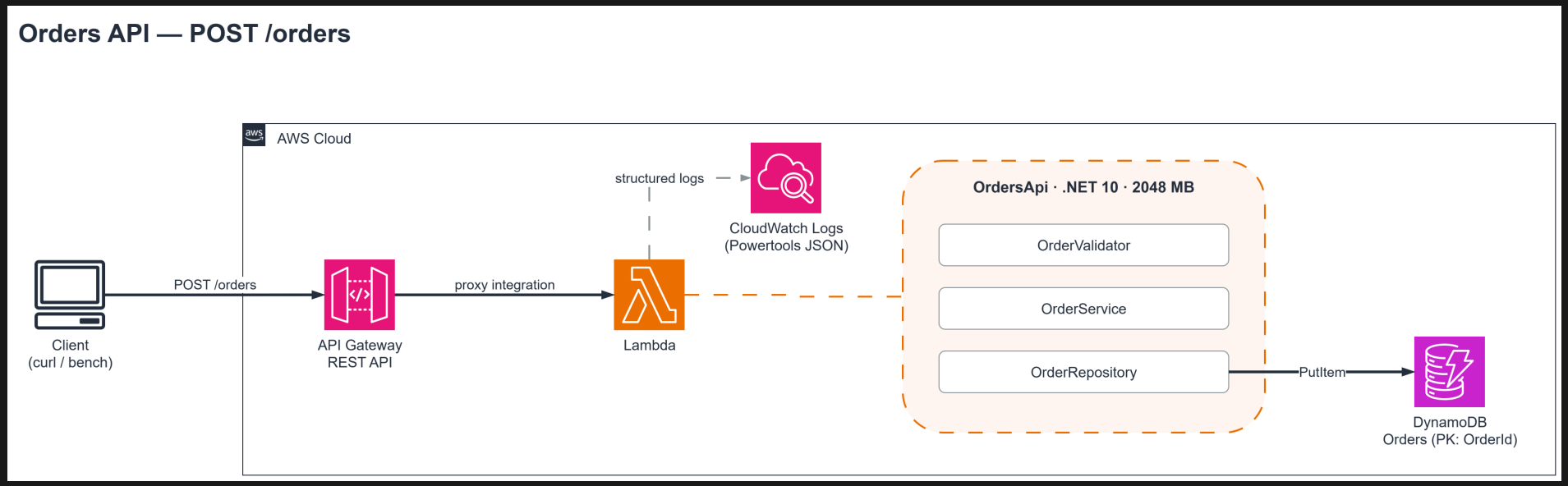
Grab the repo now



s12d.com/riv26-svs344

Every failure. Every fix.

What we're modernizing



Same code. Three stacks.



Same in all three
Runtime: dotnet10
Architecture: arm64
Memory: 2048 MB
Region: us-east-1

01-managed

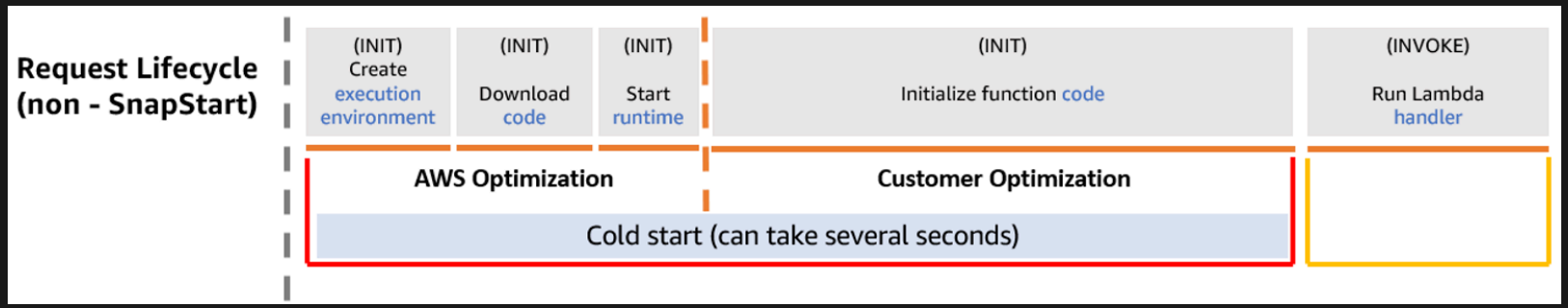
02-snapstart

03-native-aot

Managed

`01-managed/`

Where that time went

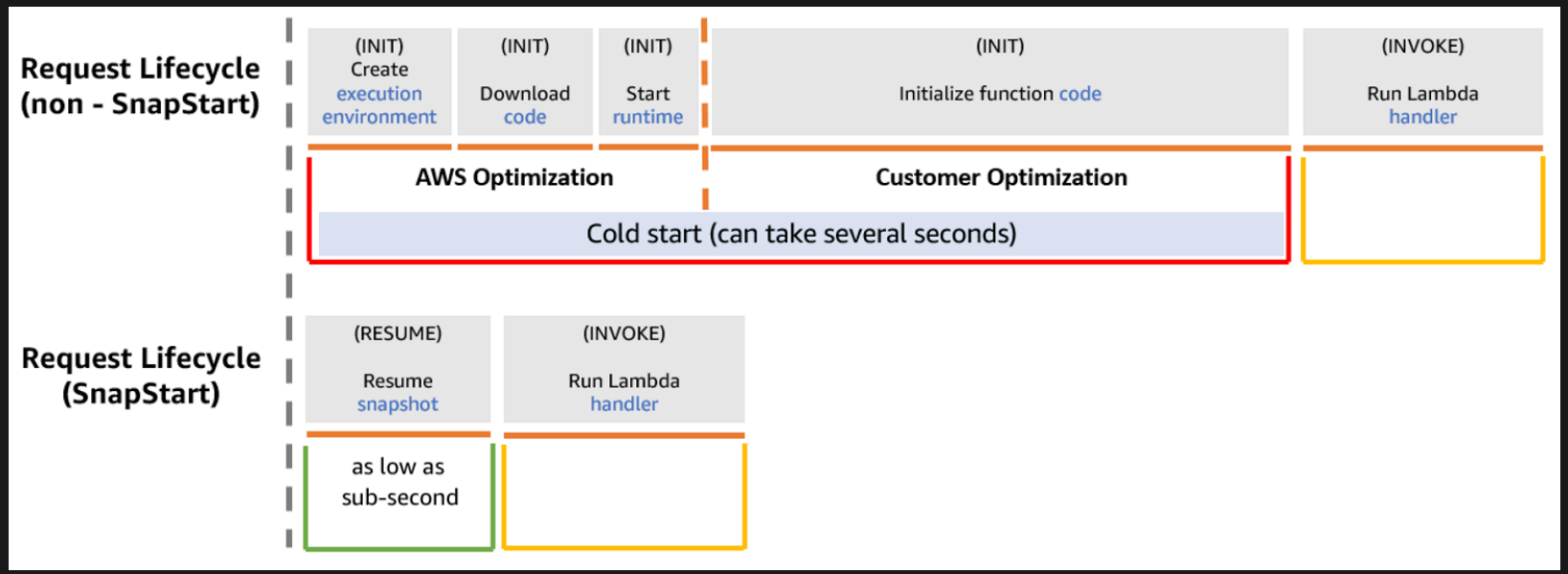


For .NET, **JIT** is the expensive part — during init, and again on the first request.

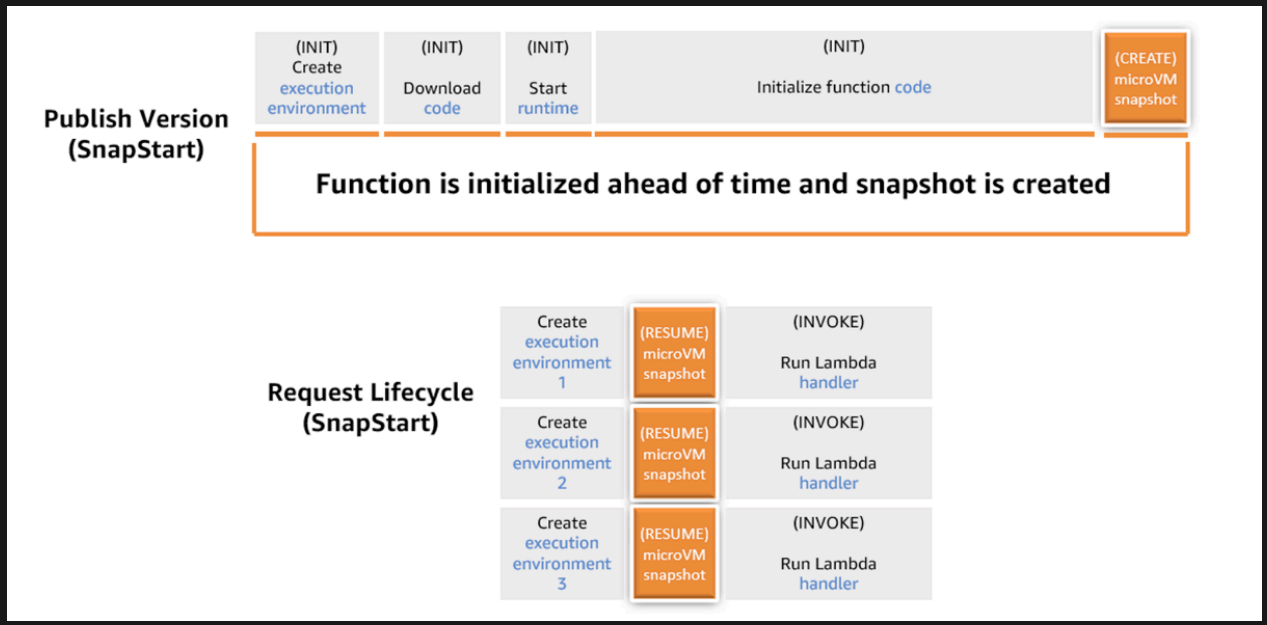
SnapStart

02-snapstart/

Same request, two lifecycles

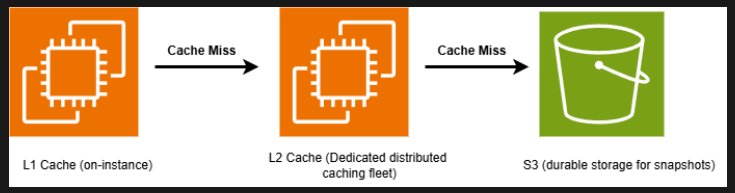


SnapStart: restore instead of init



Init runs once, at **version publish**.

Where the snapshot lives



~1ms L1 · **single-digit ms** L2 · **hundreds of ms** S3

Enabling it is not the win

	Init / restore duration (p50)	Duration (p50)	Cold total (p50)
Managed	394ms *(init)*	701ms	1097ms
SnapStart	647ms *(restore)*	1019ms	1670ms
SnapStart primed	718ms *(restore)*	**233ms**	**952ms**

Priming

Run the real path **before** the snapshot. Repeat it. Idempotent only.

Iteration	Run
1	447ms
2	8.7ms
3–10	3.4–6.0ms

Provisional — single publish. Final numbers from the bench run.

What the snapshot froze

Random seeds — re-seed after restore

Connections — release before, reconnect after

`DateTime.UtcNow` at init — never reuse as "now"

Native AOT

`03-native-aot/`

1. Serialization
2. Dependency injection
3. Trimming warnings

Head to head

	Managed	SnapStart primed	Native AOT
Cold total p50	1097ms	952ms	**320ms**
Cold total p99	1637ms	1248ms	**440ms**
Init / restore p50	394ms	718ms	**228ms**
Duration p99	828ms	309ms	**146ms**
Warm avg	11.7ms	13.1ms	**7.5ms**
Max memory	133MB	134MB	**63MB**
Effort	none	low	high

Which path

Latency-critical, willing to invest → **AOT**

EF Core or heavy reflection → **SnapStart**

Cold starts don't matter → **managed is fine**

Take these with you

Library compatibility matrix — check before you commit

Modernization patterns — the recipes, reusable

Monday morning

1. Measure your cold start. Know your baseline.
2. Enable SnapStart on one function. Prime it.
3. Pick one hot path, check the matrix, try AOT.

Questions



s12d.com/riv26-svs344

Please fill out the session survey.